

UNIT – V

Internet and Website Establishment: Introduction – Technologies for web servers – Internet tools relevant to Commerce – Internet Applications for Commerce – Internet charges – Internet Access and Architecture – Searching the Internet- Case study.

Explain Commercialization of Internet

1. INTRODUCTION

In the 1960, the U.S Department of Defense addressed the problem of creating a decentralized data communications WAN that did not have a single "Point of failure," such as a hub that could be targeted and disabled in an attack. This experiment became known as the ARPANet.

The first Interface Message Processor, a predecessor of the router, was installed at the University of California Los Angeles (UCLA) "around Labor Day in 1969".

In 1973 and 1974, a set of communication protocols for exchanging data between computers on the network emerged from the various research and educational efforts involved in this government sponsored project. This protocol set eventually became known as the TCP/IP suite. As a consequence, multiuser systems such as UNIX and VMS soon became the most popular method of accessing the internet.

In the latter part of the 1980s, the ARPANet backbone was taken over by the NSFNet, funded by the National Science Foundation. In mid- to late 1980s, government funding from the NSF, NASA and the Department of Energy made it possible for various IP-based networks the sprung up among different institutions to grow and to interconnect.

The NSFNet backbone operated as a cooperative agreement between NSF and Merit, the Michigan higher education network. Starting in the early 1990s, however, the federal investment for a general-purpose, nationwide infrastructure started to decrease and has now ceased.

COMMERCIALIZATION OF THE INTERNET

The telephone and TV companies argued in the late 1980s/early 1990s that the government ought to get out of the network business. Several carriers proposed solving the privatization problem by creating their own network with no government investment required.

The corporations and organizations see business opportunities on the Internet and are connecting to the Internet in ever increasing numbers. The Internet market providing products and services. Whether one wishes to transfer intra organizational documents or find software for a Pc, the internet contains a large number of databases and information servers located around the globe.

Users can perform searches and queries to seek such information with information tools including Gopher, WAIS, and the World Wide Web. The total amount of data on the Internet is estimated at several thousand terabytes.

However, the Internet is increasingly being viewed by the public as more than a collection of interconnected local, long-distance and international networks and the organizations hosts connected to these networks.

The following list depicts some of the activities supportable with the Internet,

- Connect to other computers on the internet with telnet.

- Copy files from and to computers on the Internet with FTP.
- Exchange ideas with other people in a public forum with Usenet.
- Search for information on the Internet with the services archive, Veronica and Web.
- Send messages to friends and associates all over the world with email.
- Traverse and search directories of information with Gopher.
- View documents, browse, search for data and traverse other resources on the Internet via the World Wide Web.

THE WEB BREAKTHROUGH

The World Wide Web is the fastest growing communications system in history. For the average individual, it is a way to jump from one Internet computer to another by clicking an underlined word or phrase called a hyperlink.

Highlights of Web:

- Proposed in 1989.
- Prototypes in 1990.
- First web servers were deployed in 1991 using line mode browsers as clients.
- First graphical user interface browsers including Mosaic from NCSA released in 1993.
- 100,000 public servers in late 1995, and recently, the number has been doubling every three months or so.
- Www is a client/server system that requires an Internet browser as client application and daemon as the server.
- Users can create, edit or browse these hypertext documents.
- The front end to the web is a point and click interface; the back end is a complex, globally linked Internet System.

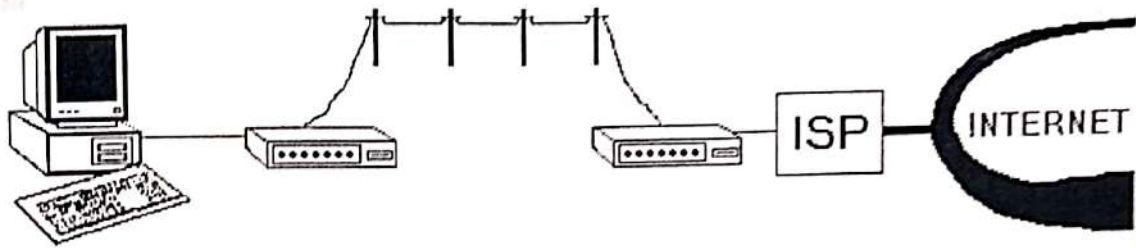
Web browser features:

- Information retrieval in the form of text with layout, images, sound, video and 3-D images.
- Hypertext
- Forms completion
- e-mail support
- date inquiry
- customization by user
- rapid integration with other applications

So the World Wide Web is a collection of clients and servers on the Internet that supports HTTP. It is an attempt to organize the information on the Internet, plus whatever local information the users want, as a set of hypertext documents.

HOW TO CONNECT TO THE INTERNET

The basic requirements are a computer, a modem, a phone line, and a contract with an Internet Service Provider (ISP).



Computer - Generally any computer purchased in the last three years, particularly a Pentium, should have no problems in supporting the software you need to install. Older machines such as a 486 with only 8Mb RAM do work, but you may want to upgrade the amount of RAM and your hard disk capacity.

Modem - This converts your digital computer information into analogue telephone signals. Choose the fastest modem that your ISP can handle.

Phone Line - This may be your biggest expense after the equipment.

Internet Service Provider (ISP) - Your ISP has a permanent connection to the Internet, and when your modem talks to their modem over the phone line, it connects you to the Internet for the duration of the phone call. The lights on your modem display show the progress of the transfer of information in both directions.

The internet is accessible from every state in the United States and from most industrialized countries in the world.

Terminal Access Services:

There are the least-expensive forms of Internet access, at an average of \$20 per month. Terminal access provides an internet user with a deal-in service to the access provider's network. Access us through a shared resource on someone else's link to the internet.

The user dials into the remote network with a terminal application package and must connect to the remote computer system through a computer account. Once the user has logged into the remote machine, the user can access the services of the internet. These connections allow a user to connect a local machine or network through a pair of modems, communicating over regular telephone lines, to the remote network of the Internet Access Provider.

Point-to-point dedicated link:

It actually places a user's computer or network on the internet. This is a costlier option, since it involves the purchase of special hardware of computer systems depending upon the type of service (speed of access).

BROWSERS

Browsers are sometimes also called web clients since they get information from a server. A browser decodes the markup symbols in web documents, turning them into formatted documents containing text and graphics. In addition browsers also originate messages that locate and retrieve documents whenever a hyperlink is selected. These links instruct the browser where on the Internet to look for a particular piece of information.

Each page that is loaded from the web is a single document, written in HTML that includes the text of the document, its structure, and any links to other documents, images and other media.

The browser communicates with the web server over the internet and retrieves a document from that server. The browser interprets the HTML markup code contained in that document and formats and displays the document. If the document contains images or links to other documents, it manages those parts as well.

Explain web server technologies in detail.

(15) MAY '10

2. TECHNOLOGIES FOR WEB SERVERS

Designing a web presentation for a company or an organization requires one to sort out the content to present, set goals, decide on topics then organize the layout and navigation of the web pages.

HTML

General features

- HTML is the markup language used to create web document.
- It is loosely a subset of the SGML
- SGML is used to describe the general structure of various kinds of documents.
- The primary focus of SGML and therefore HTML is the content of the document, not its appearance.
- Browsers decode the HTML instruction and display the document on the requester's screen

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The following is a list of basic tags for HTML

Tags	Use
<html>.....</html>	The entries HTML document
<head>.....</head>	The head or prologues of the HTML document

<body>.....</body>	All the content in the html document
<title>.....</title>	The title of the document
<center>.....</center>	Center text
<a>.....	Illustrate reference and anchor portions
.....	Make word bold face
<i>.....</i>	Italicize word
<u>.....</u>	Underline word
....	First level heading
...	Second level heading
....	Third level heading

Editor	Description
HTML Editor	It is an application that allows the insertion of tags into a file and the result is in a WYSIWYG fashion. The tags are shown in a lighter color than the surrounding text. There are option to hide the tags in the documents to depict the full affect
Microsoft Internet Assistant	It is a plug-in for Word for Windows 6.0 that allows user to create HTML files in Word and then save them as Html
Template Packages for MS word	For Microsoft word Version 2.0 or 6.0 that allows user to assign styles in a word documents and select HTML features from a toolbar

WordPerfect Internet Publisher	It is a Plug-in for word Perfect for windows .It includes a template for editing files for the web , and a converter program that support the abilities to add links and convert the documents to HTML.
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- To creating tabs a user may also create links between documents, create list, set page breaks, identify address and quotation and insert any special character and color.
- There are two ways to create home pages: do it yourself or hire a consultant who knows HTML and has the skill for the task. Clearly to create home page user must have knowledge of html.

Hypermedia

- For graphical output there are two kinds of images that a web browser can handle: inline images and external images.
- An inline images is a graphic on the web pages that does not contain a link to another place; it is there for illustration only.
- External images are not directly loaded on a web page; they are only downloaded at the request of the reader
- Inline images are specified using the tag in the HTML file
- An example of an image tag is
- However, inline images and external images provide a design and presentation for a documents that can be alluring to the customer

DATA COLLECTION

- Web server automatically create an access log of usage data, recording the domain name or IP address of site accessing the files ,dates and times of access ,which files were viewed and the sizes of those files
- A three-digit return code indicates whether file requests were fulfilled or rejected ,helping the web-master check the validity of links
- The agent file record the brand and version of browser being used; the referrer file identifies the page from which a visitor is linking and where the visitor travels within the site
- All of this information has value

PUBLISHING SYSTEMS:

There are many browsers/publishing systems that allow users to create web documents. Companies are now beginning to understand the benefits of publishing their information on the WWW.

Two of the most common publishing systems are Netscape publisher and Adobe PageMaker.

Netscape:

In the past, many publishing groups wishing to establish an electronic presence had to partner with commercial on-line services. Netscape publishing system has allowed enterprises to distribute their own publications and services on the internet by offering content providers flexible and integrated software for organizing, customizing, and delivering text, graphics, audio, or video documents to users around the world.

Netscape publishing system is an integrated solution that is easily customized and can be deployed quickly. It can be used as a standalone solution or be integrated with current software systems.

The following is a listing of some additional standards tha Netscape publisher complies with.

- Native support. Available for HTML, HTTP, FTP, NNTP, SMTP, MIME, S/MIME and POP3 standards.
- Cross-platform support. Provides a common interface and common behavior among the different platforms.
- Open Environments. Works in environment that support HTTP complaint network clients, including Winsock.

The following is a listing of support **platforms for the publisher.**

- Apple Macintosh
- Intel(x86)-based
- UNIX

Adobe PageMaker

Adobe PageMaker is a professional, cross-platform, desktop publishing program that allows the ability to design and produce sophisticated publications.

PageMaker combines text and graphics from a wide range of software applications for printed or electronic delivery. PageMaker offers tools for producing professional-quality publications. PageMaker helps produce accurate color. It allows users to separate images directly within PageMaker.

The key features are.

- Ability to print files to disk as multipage EPS files or as postscript language files.
- Ability to print reader's spreads
- Ability to save publications in PageMaker 5.0 format
- Access to more printer-specific features in the print dialog box.
- Compatibility with adobe acrobat including Create Adobe PDF

- Image manipulation using Adobe Photoshop effects filters.
- Link management controls for imported text and graphics files
- OPI reader and writer
- PageMaker scripting language for scripts that automate repetitive tasks
- Plug-in for creating portable document
- Support for Adobe postscript, PCL and quick Draw printers.
- Two-way printer communication
- User-definable thumbnails
- Work with a host of add-ons and all the leading software programs

The following is a listing of support **platforms for the PageMaker**

- Apple Macintosh
 - Intel(x86)-based
 - UNIX
- All of Microsoft's Web server technologies build upon a common foundation—Internet Information Services (IIS). Microsoft and third parties use IIS's two plug-in architectures—Common Gateway Interface (CGI) and Internet Server API (ISAPI)—to build more sophisticated Web development platforms. In some cases, Web tools require a specific combination of programming language and tools, such as ASP (which relied on scripting languages) or ATL Server (which only supports C++). But the most successful products have been those such as ASP.NET that support a variety of programming languages.

Tools	Third Party	Visual InterDev	Visual C++	Visual Studio .NET, Web Matrix	Visual Studio 2005, Web Developer Express
Languages	C++, Perl	VBScript, JScript	C++	VBScript, JScript, VB .NET, C#	VBScript, JScript, VB .NET, C#
API/Framework		ASP	ATL Server	ASP .NET 1.0	ASP .NET 2.0
Web Server	CGI	ISAPI			
	IIS				

3. Internet Tools Relevant to Commerce

LIST AND EXPLAIN THE VARIOUS INTERNET TOOLS THAT ARE AVAILABLE BESIDES WWW
(15) MAY 2010

This section describes other Internet tools that are available beside www.

Archie

As noted in the introduction of this chapter, Archie is a directory service on the Internet which indexes and stores file titles for a large number of anonymous FTP sites. There are a few dozen Archie hosts worldwide. There are three possibilities:

- Users can run the Archie program (on an Archie server) as a client.
- Users can Telnet to a nearby Archie host and use keywords.
- Users can e-mail an Archie server and receive the response by e-mail.

The program searches its lists and e-mails the results to the user, providing the user with a likely site offering FTP access. The e-mail reply file is likely to be huge if the user does not define the search well.

File Transfer Protocol

FTP is an application-layer mechanism (a high-level protocol) for file transfers that are widely used on the Internet (with TCP/IP). The protocol specification describes how one computer can interact with others for the purpose of transferring files on either direction. Users can see directories on either system and perform file management functions. A VIP site is an archive of files. Some will be available only to password-holding users, while others will be free to anyone to download and use. In general, the files are downloaded only to the nearest on-line host. Anonymous FTP means that the library of files is available for free use by anyone. One does not need to make payment or to establish a prior relationship with the organization. If asked, the name is *anonymous* (or guest) and the user id/password is the organization's network e-mail address.

The key FTP commands are:

```
ftp<remote.computer.name>  
open<remote.computer.name>  
cd (change directory)  
In (list files)  
get (transfer the files to the ftp client)  
close (hang up)
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Gopher

As noted, Gopher is an Internet service that allows the user to browse Internet resources using lists and menus. Gopher groups information resources by category this is a tree-branch approach to information searching. The user can navigate down

through a hierarchy of categories and subcategories until the information is found. Gopher is usually fast and it is often the best way to find information resources. It runs as a client on the user's PC; the user can either connect directly to a Gopher server or can initiate a Telnet session to a Gopher site. The user can navigate the Internet by selecting items from Gopher lists. Some Gopher servers will go further than just provide a user with their list information: they also search for information categories through the Internet. Anyone can set up a Gopher server and offer it over the Internet hence, there are numerous special-interest Gopher sites run by enthusiasts who are willing to help others find information.

Telnet

Telnet is a terminal emulation protocol that is associated closely with TCP/IP and used over the Internet to contact remote host computers. Telnet provides DEC VT-100 emulation as its basic emulation; however, it is more than just a simple terminal emulation protocol since it can support commands using IP. Telnet allows the user to log on to remote computers, access public files and databases, and sometimes run remote applications. It allows one TCP/IP system to emulate a character-based terminal on the other, and so it allows PCs to act as if they were directly connected through the aerial port of a remote system.

Veronica

Veronica (Very Easy Rodent-Oriented Netwide index to Computerized Archives) is an Internet menu-based facility that helps the user find the Gopher servers that may contain the information needed. It is an aid to finding the titles of files or documents. It is often available from Gopher sites. The user browses Veronica in the same way (using menus and lists) as Gopher.

WAIS

WAIS (Wide Area Information Server) is a distributed information service available to search Internet database indexes using simple natural language input. It allows the user to perform keyword searches of the full-text using electronic forms. WAIS searches for keywords within the files, not just for titles, as is the case of Archie. The search strategy is more sophisticated than Archie and Veronica; these only allow the user to search for keywords in titles. WAIS also has a feedback mechanism which enables the users to refine their search strategy progressively. WAIS requires special servers, but these are usually also made available from WWW and can also be accessed through Gopher menus.

Usenet Newsgroups

These public bulletin boards or discussion groups contain a large collection of opinions, comments, questions, and answers from Internet users everywhere. The messages are posted via e-mail. While most conferences are completely open to the public, an increasing number are moderated; that is, messages cannot be directly posted to the conference, but are instead posted to a human moderator who chooses which messages to display.

Other Internet Applications

IRC-Internet Relay Chat: IRC is like CB radio, except that instead of talking, users use a computer screen and a keyboard. From the client point of view, it is similar to PHONE which was a standard utility on DEC VAX stations and on UNIX systems. The IRC channels are like a telephone party line: many people can be on a line at the same time, everybody can hear (see) the conversation in real-time, and can either join in or just hang out. There is no limit to the number of conversations that can take place. New channels can easily be opened up. Channels can be private, secret, moderated, or invite-only and so can be used for serious business conferencing. There are IRC servers around the world.

Internet Voice Chat (Voice over the Internet): The first version of the Internet Voice Chat (IVC) shareware program became available in 1994. A direct link supporting compressed voice can be established between two computers connected to the Internet using SLIP or PPP and running the IVC software. Users must know the IP address of the person they wish to call. Although this permits two-way voice communications through the Internet at rates lower than standard long distance rates, IVC has been more of a novelty than a useful application; this is because the Internet is currently based on store-and-forward technology and so the quality is spotty at best.

4. INTERNET APPLICATIONS FOR COMMERCE

- Internet application for commerce addresses about some of the current applications of the internet.
- Internet hosts store topic papers, software and other document. Others on the internet collect this information together so that they may be accessed through a common source. These documents can be downloaded to local machines for private perusal in a matter of seconds.

- As another opportunity, many companies started to advertise on the internet in the early 1990s.
- Web home pages establish company's internet presence. This page provides a company profile and serves as a main menu for services provided by company via the internet.
- Companies often tout their web presence as an example of their technological awareness and progressiveness.
- Company logos and any other pictures can become a part of the image put forth by the home page.
- For companies that provide products and services and receive orders an internet- based order entry program may be used.
- Web pages can contain actual data entry forms as well as text and multimedia; many companies connect order entry forms directly to their product literature. Additionally, these web forms can perform searches against information in a database and return the results as temporary, system-generated web pages for the end user to view.
- As an incentive to use the internet order entry program, a company can provide documents and information for the customer using the same technology.

USES OF THE INTERNET:

One use of internet is e-commerce .other uses of internet are

- PERSONAL MESSAGING (E-MAIL):

E-mail was one of the first application on the internet. The use of e-mail is having profound effect on the way people communicate and the way to organizations operate. An e-mail message can be quickly typed and sent. Unlike the use of the telephone, it does not need the recipient to be available to take call. The e-mail can be sent to many recipients; it is a matter of record and its electronic content can be saved, edited and / or used in other documents. The-mail does not facilitate a conversation in the way that a telephone does but it can and is replacing most memos and many telephone calls within organizations and between individuals and organizations.

E-mail is not an invention of the internet. E-mail was a service available of the internet. E-mail was a service available on the internal networks of many organizations and has been provided by other public access network services. The internet however is an ideal tool for e-mail as it is the one network that can connect all users the internet is the default option for an e-mail services unless privacy requirements dictate a more secure provision

DATA INTERCHANGE:

EDI has been traditionally transmitted over proprietary VADS. EDI started before the internet was widely or commercially available and made use of either VADS or point to point connections. Users of EDI have been reluctant to transfer their communication needs to the internet because of the concerns with security and reliability. Some EDI requirements are, however, being transferred to the internet as its usage is generally cheaper than a VADS. There are also hybrid systems where EDI messages are taken in by a clearing operation, decoded and forwarded, via the internet, to small businesses user with limited EDI requirements and no EDI provision.

TELEWORKING:

It is another practice that predated the general availability of the internet. Telework has a number of definitions but it generally involves doing work that has an IT component at home and using telecommunication to communicate with that office.

Full-time teleworking has not materialized as the radical shift in working practices that was predicted by many pundits. Informal teleworking has, however, become a common practice with employees spending the odd day working from home; the general availability of internet access has been an important facilitator of this change in working practices.

DISTANCE EDUCATION:

The internet is being utilized by colleges as a facility for the delivery of distance education. The traditional vehicle of distance education has been 'print through the post', supplemented in recent years by radio, television, video and computer aided learning packages. The internet has the facility to replace all of this as a multimedia offering through a single delivery system. The internet is being utilized by traditional distance learning institution such as UK open university and is being leapt in by other institutions keen to get in on the act. The internet can be great facilitator of distance education ; To be worthwhile it still requires quality material and through support of the students.

ENTERTAINMENT:

In addition to specific uses of the internet there is a recreational use of the internet, the surfer. On the internet people can play games, find snippets of information, join a chatroom or just admire the intricacies of other web sites.

The internet can even be place to find a partner – hopefully that does not classify as an e-commerce transaction.



Early web commerce revenue has been made in four areas:

- ✓ Direct selling or marketing of a company's existing products and services,
- ✓ Selling advertising space,
- ✓ Charging fees for the actual content accessible on a web site,
- ✓ And charging fees for online transaction or links.

DIRECT SELLING

- ✚ A group of companies has begun selling their products directly on the web.
- ✚ The web reach can transfer a small company into a global distributor.
- ✚ Smaller companies and so called cyberpreneurs are moving faster and getting more substantive results.
- ✚ While no location aspect of web commerce can be advantageous, companies are also finding out that customers have a time discovering a particular site among the hundreds of thousands out there.
- ✚ As a result, more web merchants are paying a sales commission as well as an advertising fee in exchanging for prominent placement on high-traffic web sites, such as search engines and home pages of online service providers.
- ✚ For example: amazon.com



❖ SELLING AND SPACE

- ✦ Companies selling ads on the web say that this business model is a nature extension of their other lines of business.
- ✦ Internet users going to the site can view ads organized by type of work and corporate profiles. The site lists several thousand jobs.

❖ CHARGING FOR CONTENT

- ✦ Content providers charge for subscriptions and also count on advertising revenue from their sites. Observers believe that advertising drives business.
- ✦ Forrester expected companies would spend \$2.2 billion advertising on the web by year 2000.

❖ CHARGING FOR SERVICES

- ✦ Another model involves charging for some type of services such as searching databases, providing space, linking, and other services on a web site.
- ✦ Industry.net offers business a place to shop for goods. It generated \$30 million in 1995.

- ✚ The business not only users to access the data but also allows them to place orders by e-mail.

5. INTERNET CHARGES

- The internet was free in its early time and is still free in the sectors of research centers, universities/college, and government organizations.
- The regional networks, supercomputing centers, and some of connections for universities, college, and government organizations were subsidized by the government.
- For instance, NFS promoted universal educational access in the 1980s by funding campus connections if the campus had a plan to spread access around. The connections are free for these campuses: as long as uses can get connected to the Internet.
- There is no internet corporation that collects fees from all internet networks and users: every network pays for its part.
- Every users at least pays for the hardware and the connection to the internet.
- The following two sections show the ways and the cost of getting on the internet for information retrieval and information sharing.

❖ BROWSING FOR INFORMATION

- ✓ If the purpose of accessing the internet is only to browse for information, then the task is easy.
- ✓ A user needs to get the computer/terminal/server and software, choose an ISP, and obtain a suitable connection.
- ✓ An internet terminal (also called network computer) should have a simple operating system and be able to download applets and files form servers, thus supplying the functionality of a PC at a much lower cost. The user may be limited to receive images, audio, or video data.
- ✓ With a PC, the user can be connected and configured as a client on the Internet by a low- or high-speed connection. With browser software on the PC, the user can read home pages on Web servers world-wide.

- ✓ If the user wants to access audio and video recourses, the user needs to have the appropriate equipment, such as a CD drive, speakers, and speech phone on the PC.

BROWSING AND PROVIDING INFORMATION

- There are three ways to provide information as home pages on the Web: "roll your own," outsource, or create them in-house.
- **"Roll your own."** It is a low-cost, entry-level strategy to create your own World Wide Web pages and post them on the Web.
- **"Outsource."** For many businesses, working with a commercial Web services provider is a good choice. Many of them provide secure transactions. You can also create pages; get the assistance of a consultant or an ad/market agent, then lease commercial Web space.
- **"In-house development"** Businesses with the resources to hire or train the necessary personnel can set up Web servers within their businesses. They can install a Web server on their own system, set up an Internet node, and create their own multimedia documents. This provides the best control over access, design, and content. However, it is only cost-effective for large businesses.

There are many options for getting on the internet: they are

- Low speed serial line
- 56-kbps leased lines
- 56-kbps frame relay.
- T1
- ISDN
- T3

❖ SETTLEMENTS

- An important issue is that of financial settlements between internet providers.
- Their similar situation is with people's telephone services. When a telephone call is made, revenue is shared between the local networks and the long-haul network.

- More advanced tools are being developed to track web site traffic and data transactions, which help to determine the use of the internet and servers. The data the tools collect will be useful to determine a better pricing mechanism.
- The internet is worldwide and many web servers and internet hosts are outside of the United States.

**DISCUSS ON INTERNET ACCESS AND ARCHITECTURE. (15) MAY 2011 (or)
ILLUSTRATE THE SPRINT NAP ARCHITECTURE IN DETAIL. (15)NOV 2009**

6. INTERNET ACCESS AND ARCHITECTURE

Internet access services are provided by ISPs, while backbones are provided by NSPs. A typical transaction over the Internet travels over the access ISP, over one or more backbones, and over a delivery ISP.

The information is processed and/or provided by a server, and the desired output travels to the originator via the opposite path.

Internet communication in the United States is achieved by connecting the backbones at four major NAPs(network access points).

These access points were awarded by the NSF. The companies involved are

Ameritech-Chicago

Pacific bell-San Francisco

Sprint-Pennsauken.

An ISP must either connect directly to an NSP. ISPs/NSPs interconnect to one of the four key NAPs, in order to provide access service to the servers on other backbones/NSPs/ISPs.

They can also have private peering points. Routing services between the NAPs are provided by routing arbiter services.

The NAPs and routing arbiters constitute the primary components of the Internet architecture that existed in the early 1990s.

ROUTING ARBITERS

The RA is the organization that provides routing information at each NAP. The RA provides customized routing information at each NAP which reflects all bilateral agreements between the NAP's clients.

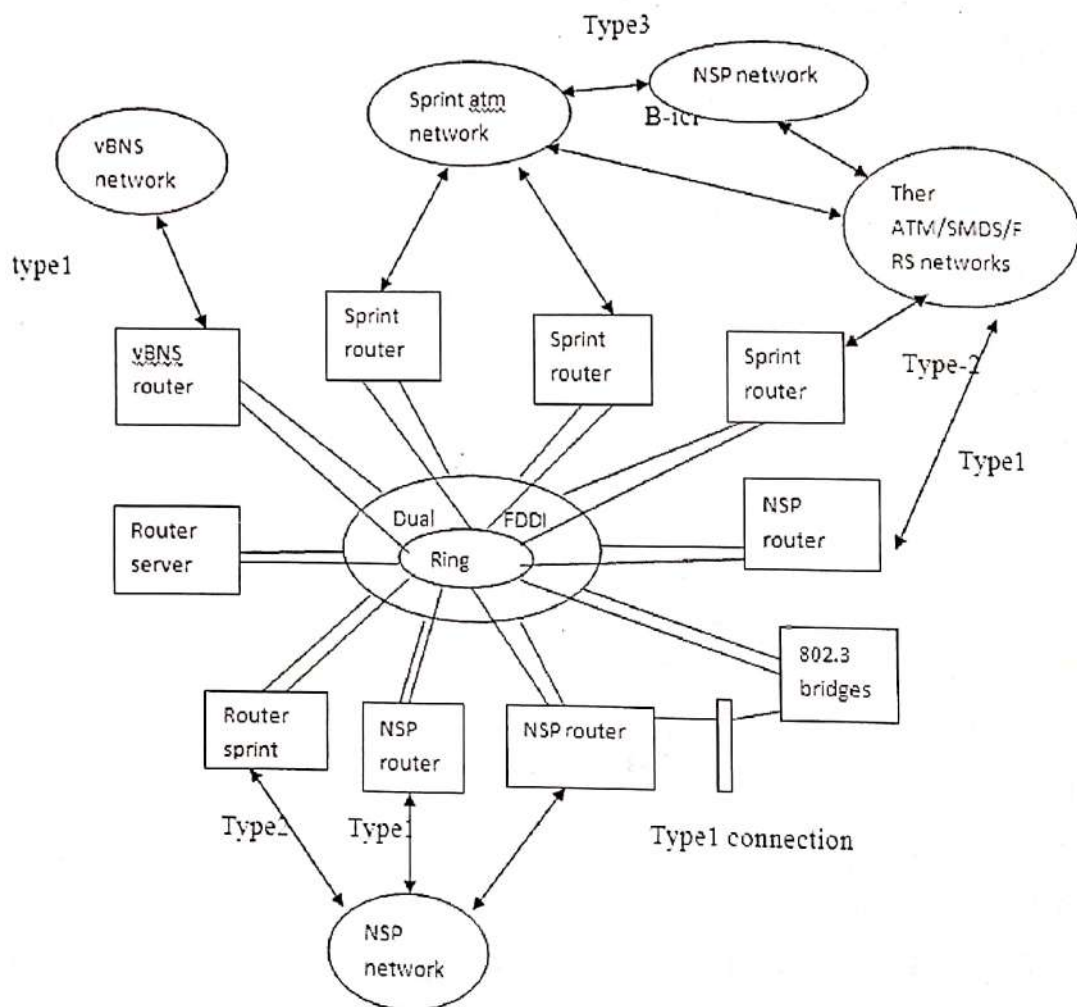
The RA is provided under award from the NSF; merit Network, and the Southern

California Information Sciences Institute, in conjunction with Cisco Systems and the University of Michigan.

The efforts of the RA are as follows:

- Route servers
- Network management system
- Routing arbiter database
- Routing engineering

EXAMPLE OF NAP ARCHITECTURE



NAP ACCESS

Sprint supports access from an NSP's router located at Sprint's physical Nap facility. The NSP can supply and maintain the router which interfaces directly with Sprint's NAP, or it can choose a Sprint-supplied router.

Sprint permits connectivity from any NSP router provided it meets the standards

for collection on sprint's NAP via a serial link or a switched service from dispersed locations using any type of WAN technology the NSP choose to use. Sprint accepts and adapts to any reasonable means of WAN access technology to the NAP.

WAN technology supported includes the following:

Dedicated circuits – DS1 and DS3

Switched services – ATM, frame, relay, and Switched Multi-Megabit Data Service (SMDS)

Sprint has established three options for NSP connectivity:

- Type I: NSPs terminate a connection in an NSP-provided router collocated at a Sprint NAP.
- Type II: NSPs terminate a connection in a Sprint-provided router at the Sprint NAP.
- Type III: NSPs terminate a network connection using Sprint's ATM user to Network Interface (UNI) at the Sprint NAP.

The Sprint NAP may also be accessed by any one of the three switched services:

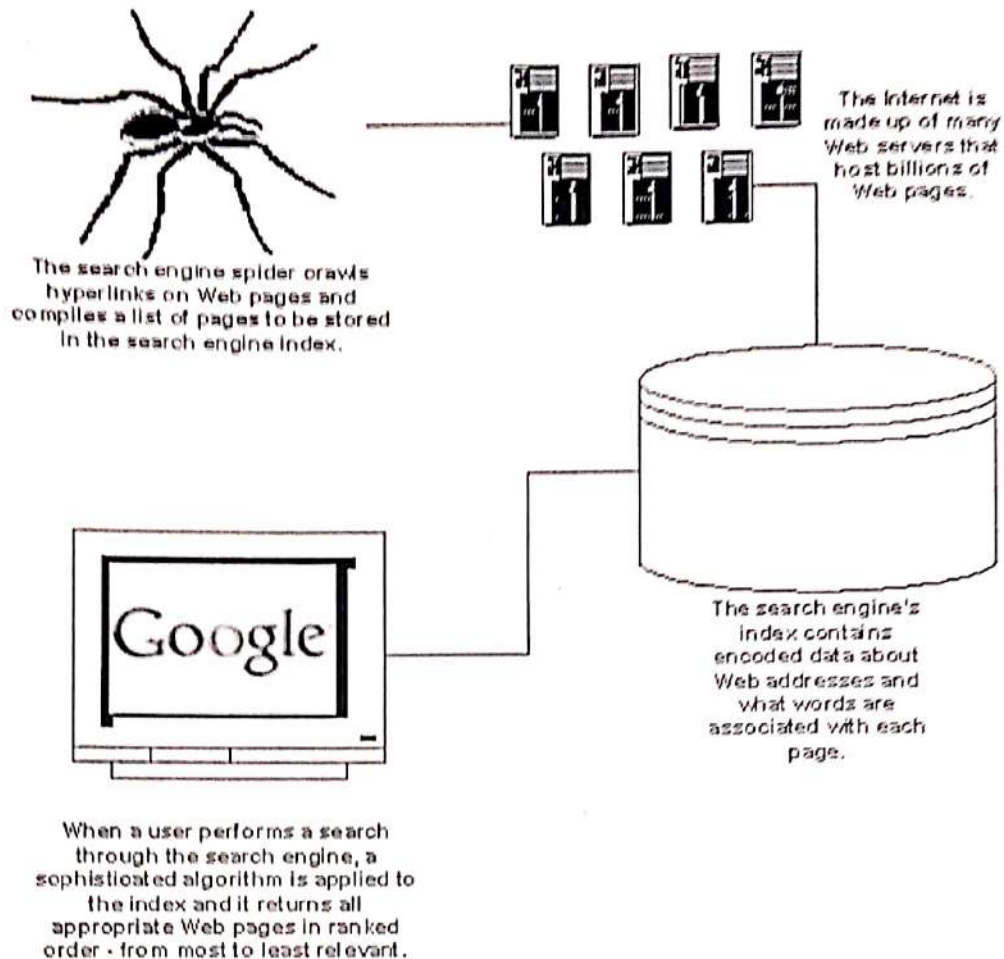
- ATM
- FRAME
- RELAY OR SMDA

HOW DO YOU LOOK FOR REQUIRED INFORMATION IN INTERNET? EXPLAIN. MAY 2011 (15) / MAY 2010 (8)

7. SEARCHING THE INTERNET

As the amount of content on-line increases, it becomes more and more difficult for users to find what they are looking for on the Internet.

❖ GATHERING INFORMATION: SPIDERS AND SEARCH ENGINES



- Most users use one of the many search engines readily available on the Internet. Commercial search engines such as Lexis (legal profession search engine) and Nexis (news retrieval service), both provided by Mead Data Central, have existed for years before the current popularity of the Internet.
- With the continuing evolution of the Internet, however, specific search engines geared toward the Web and newsgroups are being established- Lycos, Webcrawler, and Magellan, to name a few.

In most instances, the term search engine is used to define the entire search service and process on a database provided by a particular vendor. Actually, the search engine is just a part of the process. The process consists of three main components:

- Sourcing information
 - Index, catalog, or database creation
 - The search engine
- The entire process will be called the search process, while search engine implies only that piece of the process.
 - Most Internet search processes fall into two groups based upon

how the data-base or index is compiled: meta-index or spider-generated index.

- In a meta-index system, the administrator of a search process manually constructs a database of documents containing pointers to various Web or newsgroup resources.
- This process can be labor-intensive and hard to maintain, considering the growth the Internet is experiencing.
- Spider-generated index systems provide much the same services as a meta-index system but does so through a program called a SPIDER.

❖ USES OF SPIDERS

- Link validation
- HTML validation
- "What's new" monitoring
- Indexing

❖ SPIDERS AND INTERNET TRAFFIC

- While spiders may reduce user traffic on the internet, they create traffic of their own.
- In developing databases spiders, using parallel retrieval, may operate nonstop over prolonged periods of time, thus using large amounts of bandwidth.
- Spiders also place a burden on network servers.
- To help alleviate some of the flow of traffic caused by information gathering spiders, the IF-Modified-Since mechanism in HTTP may be used.

❖ LIMITATIONS OF SPIDERS

- ✓ The main limitation is that there is just too much information available through the internet and the amount is growing all the time. Although spider-generated databases are generally more current in totality than manually compiled databases, the fact that new information is being added every day precludes the spider-generated databases from being completely up-to-date.
- ✓ Aggravating the problem of processing redundant information is the basic construct of the web that puts every file at equal importance.
- ✓ Spider constructed databases may have multiple listings for the same information area.
- ✓ To address redundant and irrelevant database listings developed by

spiders, a standard for spider exclusion has been developed. Some spiders specialize only in certain areas.

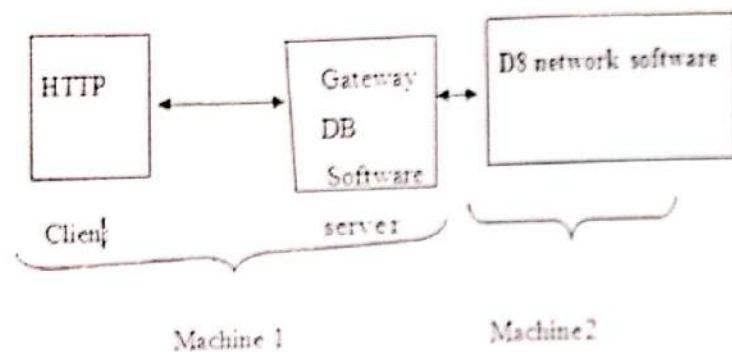
SEARCH ENGINES

➤ Search engine overview:

A search engine is a program that searches through a database. In the context of the internet, search engine is most often equated to search forms that request programs to look through databases of html documents gathered by spiders or through manual data gathering processes.

➤ Accessing the information database:

- An HTTP server provides a method of running databases searches from within an HTML document. This method is called Common Gateway Interface(CGI)
- The server can pass information to the CGI program and get back any results returned by the database search.



- CGI programs are the most common way of getting information from a database via the Internet.
- CGI programs just need to be able to either read data from user input or access certain environment variables for the data. Once the data is read, the program executes and generates results in HTML.
- CGI programs can be of many types, but the common CGI programs seem to be in Perl. Perl's text processing abilities and versatility seem to make it a good choice for CGI scripts.

➤ High-level details of using CGI and gateway software.

Processing software is called directly from the HTML search document by the HTTP client. The processing software reads the query input passed to it via the CGI call.

➤ Multithreaded queries:

Multithreaded queries are performed by using a gateway program which conforms to CGI. The program work with the HTML form describing the various search engines and type of queries that can performed.

➤ Search engines:

Search engines are similar to spiders in that they go out autonomously and seek meaningful information. Search agents, however, are confined to a single database or information source.

➤ Key search features:

✓ **Free-text search**

The ability of the search engine to accept single words, phrases, or sentences. Common words are ignored, and the documents are elected based on the words in the query.

✓ **Automatic morphology**

Nouns and verbs in English occur in many tenses; the search process automatically generates the various forms of the words and uses them in the query.

✓ **Word indexing**

Processing of collections of documents by building indexes which contain the location of every instance of every word. It is the indexes, not the documents, that are searched.

✓ **Lexical affinity search**

Search on the words that occur close to the text form of the query.

✓ **Ranking and relevance scoring**

Using a ranking algorithm the search engine assigns every document that contains at least one of the queried words a numerical score. The more often the document contains the queried words, the higher the score.

These items do not by themselves guarantee a good search engine but assist users in better defining their informational requirements

State and discuss any seven specific search tools available. (7) NOV 2009 (or)

Explain any ten special search tools. (10)NOV '10

➤ ACTUAL SEARCH TOOLS

- ✓ **Yahoo and Google:** These are the currently considered the most complete directory resources on the World Wide Web.

It is a categorized, menu-driven directory of web resources. User can also execute an automatic search on their databases from their main pages.

General search engine's data bases are created by automatic robots which travel the web on a regular basis.

Hence, their databases are larger than yahoo's, but it also means that they are not intelligently categorized. Thus, the users can only search for certain terms which appear on the page, in the title, or in manual keyword fields.

- ✓ **infoSeek :** This search engine accepts natural language questions, such as "What is a good restaurant in new York city that serves fresh morels? But also provide keyword search capability.
- ✓ **AltaVista:** This is a search engine that is very likable
- ✓ **Open Text:** This is a technically superior search engine on the internet and produces fast, relevant responses.
- ✓ **WebCrawler:** It is faster than Infoseek and lets the user view more matches on the screen simultaneously. It does not screen its matches for relevance as much as infoseek does, hence users may find matches that Infoseek omitted
- ✓ **Lycos:** The Lycos robot indexes not only the sites that it visits, but also every link on those sites.
- ✓ **Savvy Search:** One can search all of the previously mentioned databases simultaneously using a tool called SavvySearch. It is not a search engine itself- rather, it is a client that executes searches on several engines (or more) in parallel.
- ✓ **Galaxy:** It searches for resources on the web, including Gopher and Telnet.
- ✓ **Usenet:** If you are searching for something current, then Usenet is an excellent resource. You can read news directly or search a Usenet archive such DejaNews for what you are looking for. Network news is organized into newsgroups; it is easy to work through the major heading and then through the newsgroups themselves. Usenet is not a network and Usenet does not even need the Internet. Rather, what drives Usenet is akin to an

agreement set up between those who want to distribute and those who want to read newsgroups.

- ✓ **Archie** : If you are searching for computer software, then Archie, is an excellent resource for navigating through the content on anonymous FTP archives throughout the world.
- ✓ **Gopher**: Gopher is service that allows users to view and obtain files, programs, and software. When accessing Gopher information with a graphic browser, users are presented with a series of menu choices, much like the directory structure one sees in the MS windows file manager.
 - ✓ By double clicking on icons users can traverse the directory structure a level at a time, until they find an item of interest. The advantage of Gopher is that a user does not have to know what the item is or even where it is (unlike FTP) If the item is a file to download, gopher involves an FTP session; if the item is a link to another computer. If the item is a link to another computer;
- ✓ **Veronica**: It is a self updating database of Gopher documents. It allows users to search all of the Gopher sites in the world. By entering a word or words, directories, programs, and articles with those words in them will show up in a menu for users to browse. For purpose of a Veronica search, the title is the name of the resource as listed on its home Gopher server
- ✓ **FTP**: Anonymous FTP permits users to access remote systems without actually having user accounts on the systems. Users can FTP to any anonymous site in the world using the hypertext interface. For eg., comprehensive guides to educators mailing lists and Usenet newsgroups are available
- ✓ The CERT archive contains items relating to computer security, warnings and announcements, security and network utility programs and papers on viruses and computer security
- ✓ **Telnet** : Telnet allows users to remotely log into other computers and can give them access to databases and many information services. This allows users to log in remotely to hundreds of worldwide libraries, campus wide information services. These sites mostly contain archives of audiovisual resources and related information.

- ✓ **WAIS :** It lets the users search indexed information to find articles containing groups of chosen words. WAIS is like Gopher in that it shields users from having to know on which computer the information resides; but unlike Gopher, WAIS does the searching for the user. A WAIS search is not totally accomplished without some human intervention. Someone has to make the text information available on a WAIS server by indexing it. But once that happens, anyone can gain access to it. From the WAIS client, users must decide which library of information to search and which words to search on . WAIS then returns articles and documents containing those keywords.
- ✓ **Internet Directory Services.:** There are a number of ways to create resources that provide access to e-mail addresses of individuals and institutions and contact information for internet services. Directories to internet addresses for people and services indexed by name are often referred to as white pages, while those directories that allow access to addresses of services by category are usually called yellow pages.

➤ **BUSINESS AND FINANCE SITES.**

- ✓ FINweb-financial economics.
- ✓ NETworth
- ✓ NASDAQ financial executive journal
- ✓ Computer science sites
- ✓ Health care and medical databases.

The Status of E-Commerce in India and its Potential
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INTRODUCTION:

E-commerce stands for electronic commerce and pertains to trading in goods and services through the electronic medium, i.e. the Internet or phone. On the Internet, it pertains to a website, which sells products or services directly from the site using a shopping cart or shopping basket system and allows credit card payments.

E-COMMERCE IN INDIA TODAY

Today E-commerce is a byword in Indian society and it has become an integral part of our daily life. There are websites providing any number of goods and services. Then there are those, which provide a specific product along with its allied services.

A) MULTI-PRODUCT E-COMMERCE:

Some Internet portals provide almost all categories of goods and services in a single site; hence, they are targeting buyers of every possible product/service. The most popular examples are www.indiaplaza.com, www.thebestofindia.com, www.khoj.com, www.sify.com, www.rediff.com, www.indiatimes.com and so on.

These Indian E-commerce portals provide goods and services in a variety of categories. To name a few:

1. Apparel and accessories for men and women
2. Health and beauty products
3. Books and magazines
4. Computers and peripherals
5. Vehicles
6. Collectibles
7. Software
8. Consumer electronics
9. Household appliances
10. Jewelry
11. Audio/video entertainment goods
12. Gift articles
13. Real estate and services
14. Business opportunities
15. Employment
16. Travel tickets
17. Matrimony
18. Pets...and more.

B) SINGLE-PRODUCT E-COMMERCE:

Some Indian portals/websites deal in a specialized field, for example:

1) AUTOMOBILES:

Examples are: <http://www.indiacar.com/> and <http://www.automartindia.com/>.

On these sites we can buy and sell four-wheelers and two-wheelers, new as well as used vehicles, online. Some of the services they provide are:

- Car research and reviews

- Online evaluation
- Technical specifications
- Vehicle Insurance
- Vehicle Finance
- Dealer Locator
- Regional Transport Office regulations
- Expertspeak
- Message board...and more.

2) STOCKS AND SHARES AND E-COMMERCE:

In India today, we can even deal in stocks and shares through e-commerce. Some of the sites are: <http://www.equitymaster.com/> and www.5paisa.com/.

Some of the services offered to registered members are:

- Online buying/selling of stocks and shares;
- Market analysis and research
- Company information
- Comparison of companies
- Research on Equity and Mutual Funds
- Tracking Market Trends
- Hotline for advice on Risk Management
- 24-Hour helpdesk....and more.

3) REAL ESTATE AND E-COMMERCE:

Portals like <http://www.indiaproperties.com/> facilitate online dealing in real estate. They offer either outright purchase or lease of a property through their portal. They provide information on new properties as well as properties for resale. One can deal directly with developers/builders or through consultant/brokers.

Allied services:

- Housing Finance
- Insurance companies
- Architects & Interior Designers
- NRI services
- Property Management Consultants
- Packers & Movers
- Security & Maintenance Services
- Vaastu or Feng Shui Consultants...and more.

4) TRAVEL & TOURISM AND E-COMMERCE:

India has a rich history and heritage and e-commerce is instrumental, to a large extent, in selling India as a product, encouraging Indians as well as foreigners to see its multifaceted culture and beauty. A major Government of India portal, <http://www.tourisminindia.com/>, has a vast variety of information for a potential tourist.

The tourist destination sites are categorized according to themes like:

- Adventure - trekking, mountain climbing etc
- Eco-Themes pertains to jungles, flora and fauna
- Beaches of India
- Architectural attractions
- Forts and Palaces
- Buddhist attractions
- Hill resorts
- Desert treks
- Pilgrimage sites

Allied services offered are:

1. Passport & visa
2. Travel & accommodation information
3. Weather information
4. Festival & fair dates
5. Shopping
6. Tour Operators
7. Information on Cuisine & Restaurants
8. Car rental services...and more.

There are also sites that highlight the tourist destinations of a specific region in India, like <http://www.incredibleindia.org/>, which covers North East India.

5) GIFTS AND E-COMMERCE:

In the bygone days, one had to plan what to gift a loved one, trudge across to your favorite shop, and browse for hours before purchasing a gift. Today there are specific Indian websites making the act of gifting quick and easy to suit one's lifestyle. One such site is <http://www.indiangiftsportal.com/>.

The gifts are categorized as:

1. Collectibles like paintings and sculptures
2. Luxury items like leather goods, perfumes, jewelry boxes, etc
3. Household curios and carpets, etc
4. Toys & games
5. Chocolates
6. Flowers

7. Wood-craft & metal-craft
8. Idols for worship...and more.

Also the items can be searched by different regions of India, as every region has a special style in making handicrafts.

6) HOBBIES AND E-COMMERCE:

The most popular hobbies from time immemorial are reading, music and films. On the Indian website <http://www.firstandsecond.com/> one can buy more than 300,000 titles of books, cassettes, VCDs and DVDs.

The books cover a wide range of topics like Business, Art, Cookery, Engineering, Children's Stories, Health, Medicine, Biographies, Horror, Home & Garden, etc.

As for music and videos, they are available in English as well as in Indian languages to cater to the varied tastes and the topics range from devotional songs, old-time favorites and retro and jazz to the latest pop, rap, etc.

7) MATRIMONY AND E-COMMERCE:

It is said that marriages are made in heaven, but in the world of E-commerce they are made on marriage portals like <http://www.jeevansathi.com/> (meaning life-partner.com in Hindi) and <http://www.shaadi.com/> (meaning marriage.com in Hindi).

One can search for a suitable match on their websites by region of residence (India or abroad), religion or caste. Once registered with them, they have email facility and chat rooms too, so that the couple gets to know more about each other before making the biggest decision of their lives.

Allied services for registered members:

1. Astrological services
2. Information on Customs and Rituals
3. Legal issues
4. Health & Beauty
5. Fashion & Style
6. Wedding Planners
7. Honeymoon Destinations...and more.

8) EMPLOYMENT AND E-COMMERCE:

Two major portals like www.Monsterindia.com and www.naukri.com (meaning job.com in Hindi) are instrumental in providing job seekers with suitable employment at the click of a mouse. They have directories categorized under the headings Employers and Job Seekers. The service for job seekers is free and for Employers they charge a nominal fee. Jobs are available online in fields ranging from secretarial to software development, and from real estate to education. The opportunities offered are unlimited.

FACILITATORS OF E-COMMERCE IN INDIA:

1. INFORMATION DIRECTORIES:

Portals like <http://www.trade-india.com/> and <http://trade.indiamart.com/> maintain directories giving trade details on almost any topic, whether it is from apparel to toys, from gems to heavy machinery, or from food to employment. The products and services are listed with appropriate sub-headings to make it easy for a serious information-seeker to find what he wants. Some other similar sites are <http://www.indiagrid.com/>, <http://indiainfo.com/>, and <http://in.indiatimes.com/>.

Allied services provided by them: Message boards, chat rooms, forums, etc.

2. BANKS:

a. Netbanking/phone banking:

This is an online banking facility available for savings account holders as well as current account holders. Some of the special Netbanking services are:

- Demat accounts for sale/purchase of stocks and shares
- Foreign Exchange services
- Direct/Instant payment of bills on the account-holder's behalf
- Financial Planning & advice
- Electronic Funds Transfer
- Loans to account-holders
- NRI services...and more.

b. Credit/Debit Cards:

Banks facilitate E-commerce by providing the most vital trade instrument, namely the Credit or Debit Card, without which E-commerce would be impossible. Some of the major Indian players in this field are:

<http://www.hdfcbank.com/>, <http://www.icicibank.com/> and <http://www.statebankofindia.com/>.

POTENTIAL OF E-COMMERCE IN INDIA:

Microsoft Research India was inaugurated in January 2005 in Bangalore, as the sixth MSR laboratory in the world. The arrival of a giant like Microsoft itself speaks of India's potential as a world-player in the field of Information Technology and e-commerce; and others will soon follow. Indian researchers and technicians can further the fast emerging world of E-commerce in India and take it to greater heights.